Machine Learning – Course Notes

Beginner-friendly study material for uploading as course material.

1. Introduction to Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions.

Basic workflow: Data → Preprocessing → Training → Testing → Evaluation → Prediction

2. Types of Machine Learning

Type	Meaning	Example
Supervised Learning	Learns from labelled data.	House-price prediction
Unsupervised Learning	Finds patterns in unlabelled data.	Customer clustering
Reinforcement Learning	Learns using rewards and penalties.	Game-playing agent

3. Data Preprocessing

Data preprocessing prepares raw data for machine learning.

Common steps include handling missing values, treating outliers, encoding categorical data, scaling numerical features, and splitting data into training and testing sets.

4. Classification

Classification predicts a category such as spam/not spam or pass/fail. Common algorithms include Logistic Regression, KNN, Decision Trees, Random Forest, and SVM.

5. Regression

Regression predicts a numerical value such as house price, temperature, sales, or marks. Linear Regression is a common regression algorithm.

6. Clustering

Clustering is an unsupervised-learning technique that groups similar data points. K-Means is a commonly used clustering algorithm.

7. Model Evaluation

Classification can be evaluated using accuracy, precision, recall, and F1-score. A confusion matrix contains True Positive, True Negative, False Positive, and False Negative results.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision = TP / (TP + FP)

Recall = TP / (TP + FN)

8. Overfitting and Underfitting

Overfitting occurs when a model learns the training data too closely and performs poorly on new data. Underfitting occurs when a model is too simple to capture important patterns.

9. Cross-Validation

Cross-validation evaluates a model using multiple train-validation splits. In k-fold cross-validation, the data is divided into k parts and each part is used as validation data once.

10. Machine Learning Workflow

Collect Data → Clean Data → Explore Data → Prepare Features → Split Data → Train Model → Evaluate Model → Make Predictions

Sample Course Material

This PDF can be uploaded under the Machine Learning course in the teacher dashboard to test the Course Material Upload feature.